LAB 3 / DESIGN DECISION

Coffee Kiosk Architecture Justification

Venkatesh Panda | PES1UGCS524

Architecture selection

We chose layered architecture for the Self-Service Coffee Kiosk System. The presentation layer captures touch input, the business layer coordinates orders and the integration layer owns payment, menu data and printing.

Style analysis

Layered architecture separates touch interactions, order rules and integrations within a kiosk deployment. Microservices would allow separate scaling and deployment but introduce network calls and operational overhead for a small kiosk. Client-server would centralize menu updates across kiosks but make checkout more dependent on network and server availability.

Two reasons for the choice

1. The Touchscreen UI can change how customers select a drink and size while Order Manager continues to apply the same menu prices and checkout rules.

2. Separate payment and receipt adapters isolate vendor-specific terminal and printer commands. The Order Manager works with approval results and receipt data, so hardware integration can change independently.

Security advantage

Payment Service delegates card entry to the terminal. Order Manager receives an approval status and transaction reference, keeping full card numbers and security codes out of kiosk order records and receipts.

Performance benefit

Touchscreen UI, Order Manager and local menu queries run on the kiosk, keeping selection and review responsive while external communication is limited to payment and printer I/O. This helps meet the under-60-second checkout target.